

ANALYZE RETIREMENT PLANS WITH AN ECONOMIC SCENARIO GENERATOR

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RETIREMENT & LONGEVITY RISK

Goal: Minimize longevity risk

- Longevity Risk: The probability of running out of assets in retirement

Financial Unknowns:

- Cost of Living (inflation)
- Equity Returns
- Medical Costs
- End of Life Costs
- Age of Death

HOW IS THIS RISK QUANTIFIED?

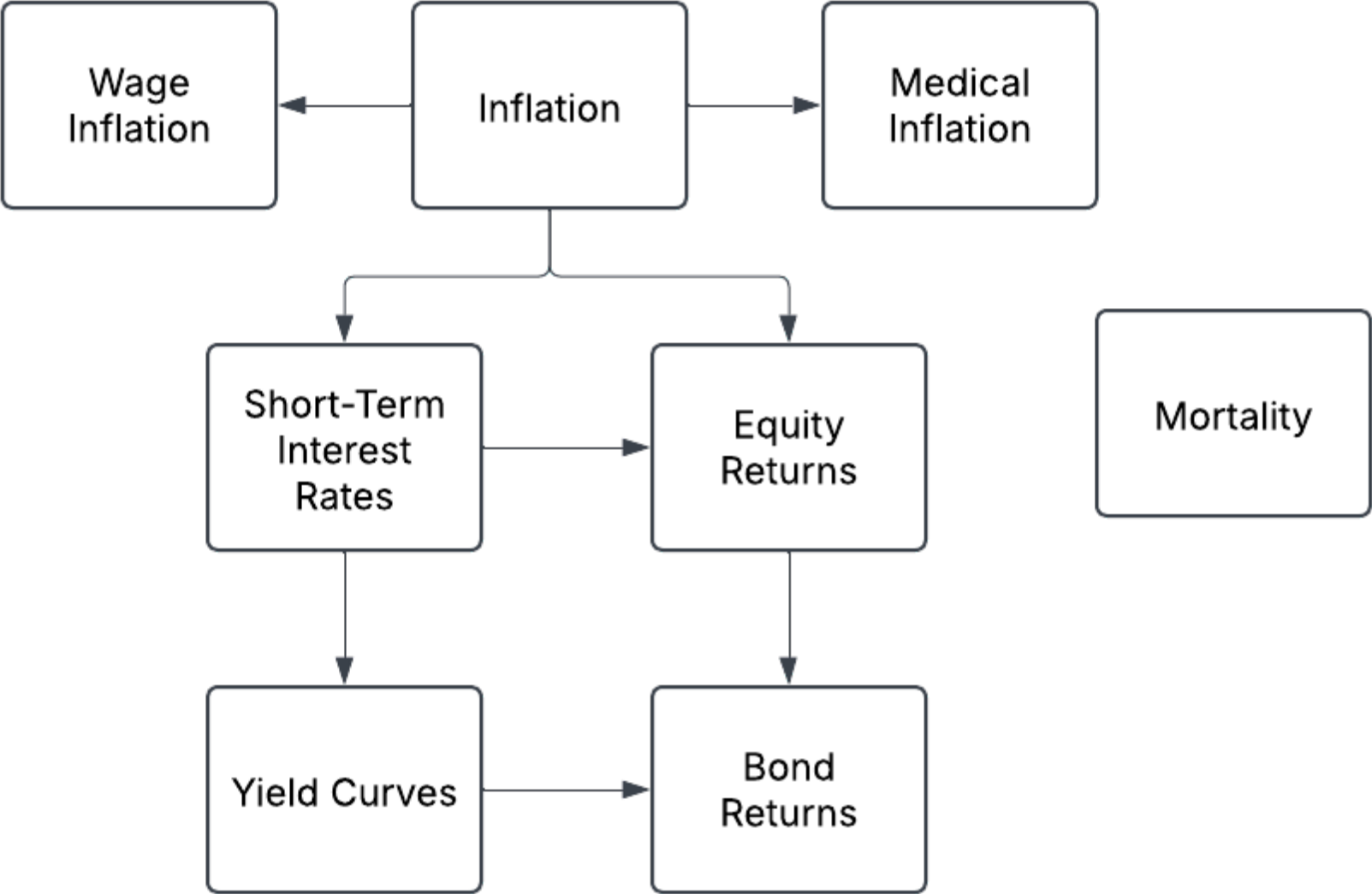
SIMULATION FROM THE PAST:

- Apply a specific “retirement plan” to a year from the past
 1. Pick a “plan” (i.e. 4% Withdrawal Rate, 50% equity portfolio)
 2. Apply that plan to a year from the past
 3. Simulate cash-flows using past data
 4. Iterate over “all” years

OUR APPROACH:

1. Build a stochastic model for different economic variables
2. Combine individual models to represent “an economy”
(Economic Scenario Generator)
3. Simulate future economic scenarios
4. Apply retirement plans to these scenarios and analyze outcomes

ECONOMIC SCENARIO GENERATOR (ESG)



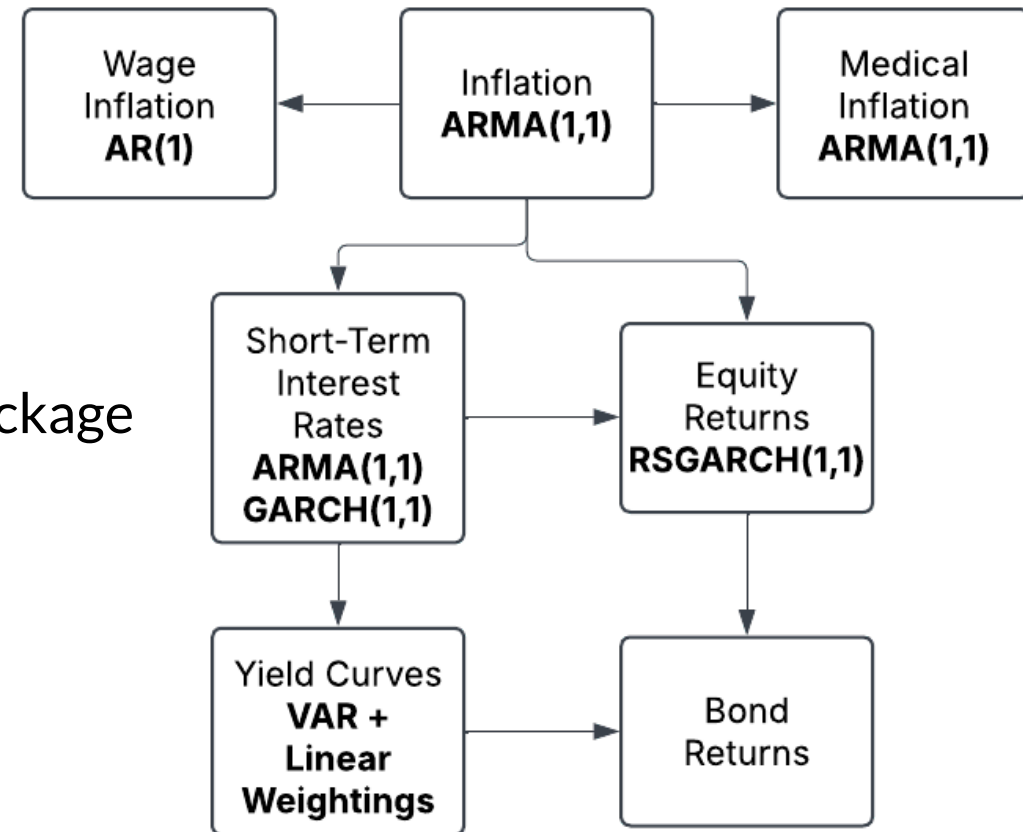
FITTED ESG

Model Selection Criteria:

1. Behavior of simulated draws
2. Model fit
3. Model parsimony

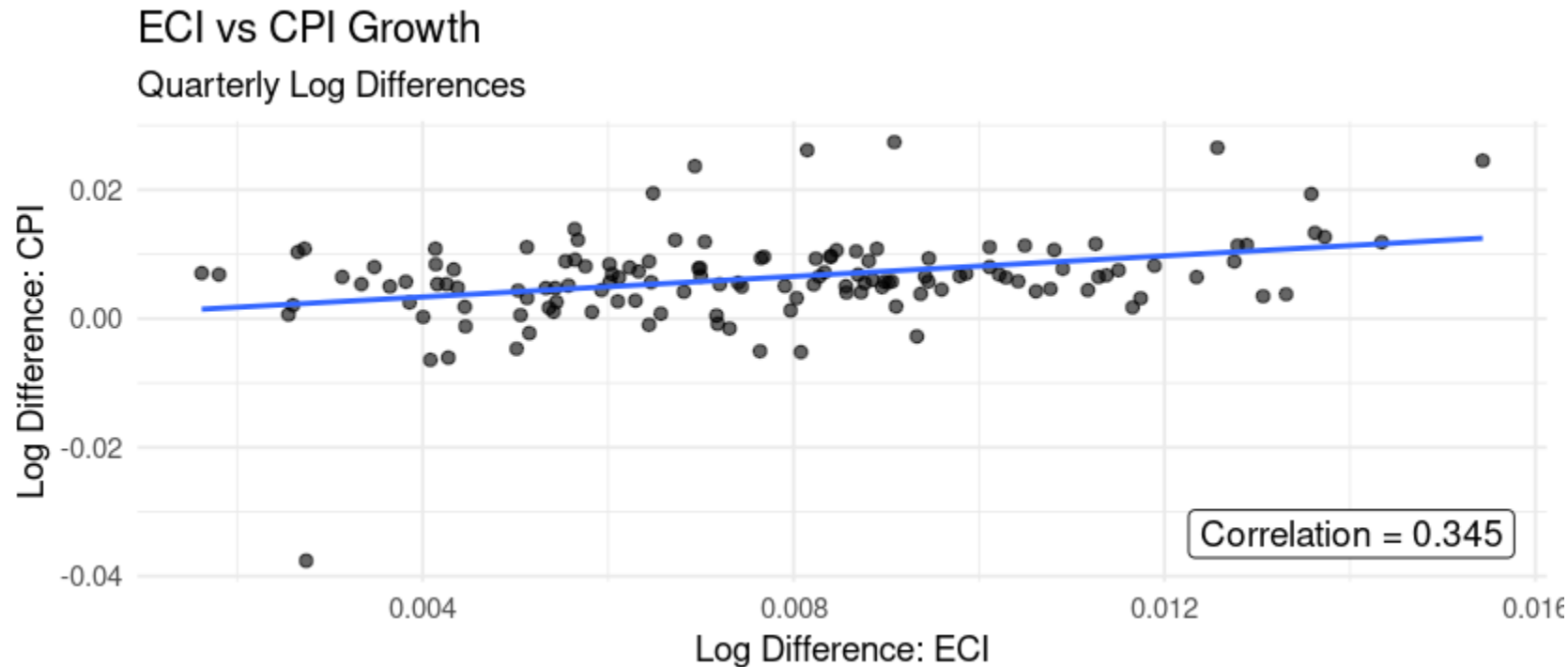
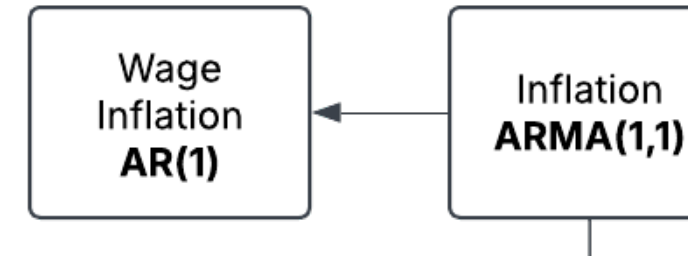
Data:

- Inflation and Interest Rates: FRED
- Equity Returns: S&P 500 from [tidyquant](#) package
- Mortality: SOA Mortality Table



WAGE INFLATION MODEL SELECTION

1. Explore Relationship between CPI and ECI

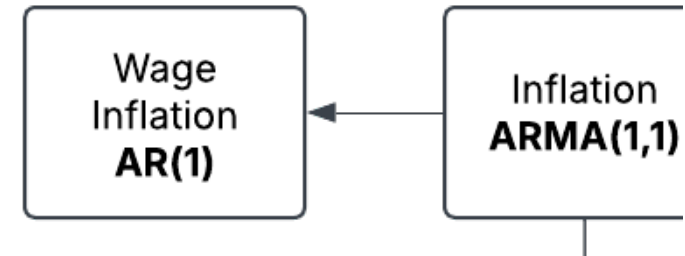


- Lagging: $Cor(ECI_t, CPI_{t-1}) = .557$

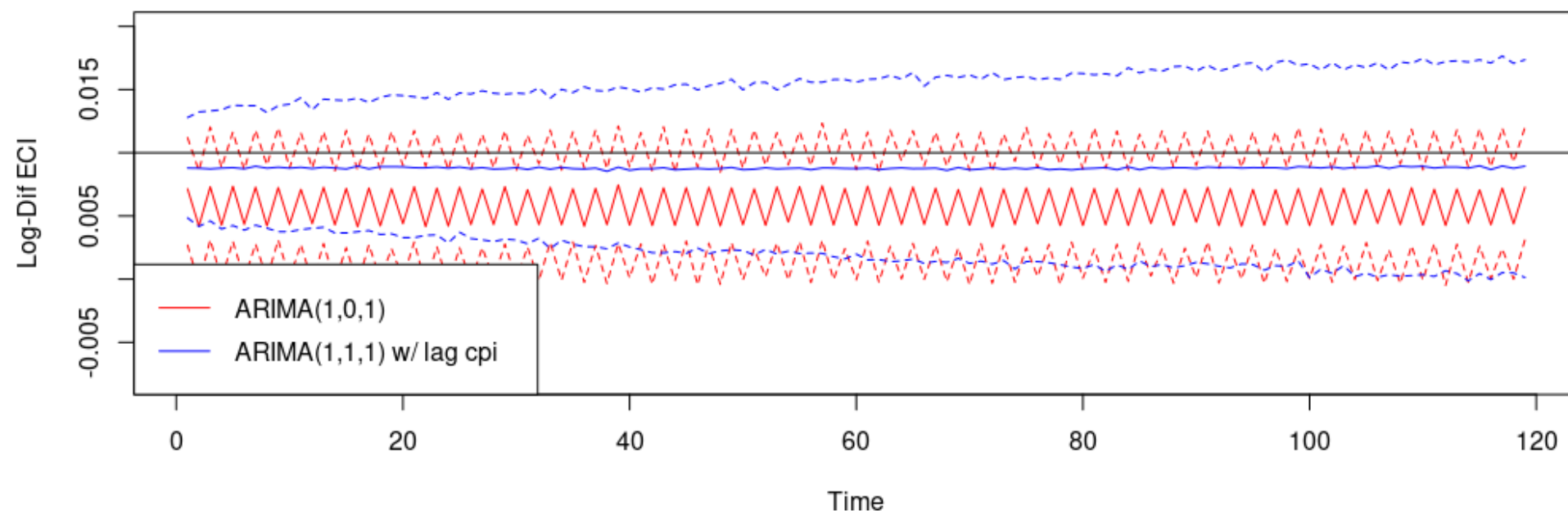
WAGE INFLATION MODEL SELECTION

2. Test Different Models

- Differing lags (for both ECI and CPI)
- Differing data subsets (1990 vs. 2010)
- Include Differencing?

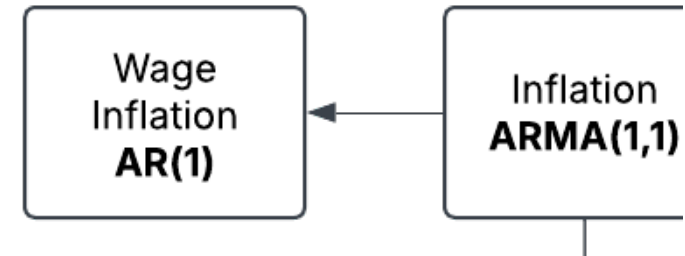


Simulated Draws for Different Models

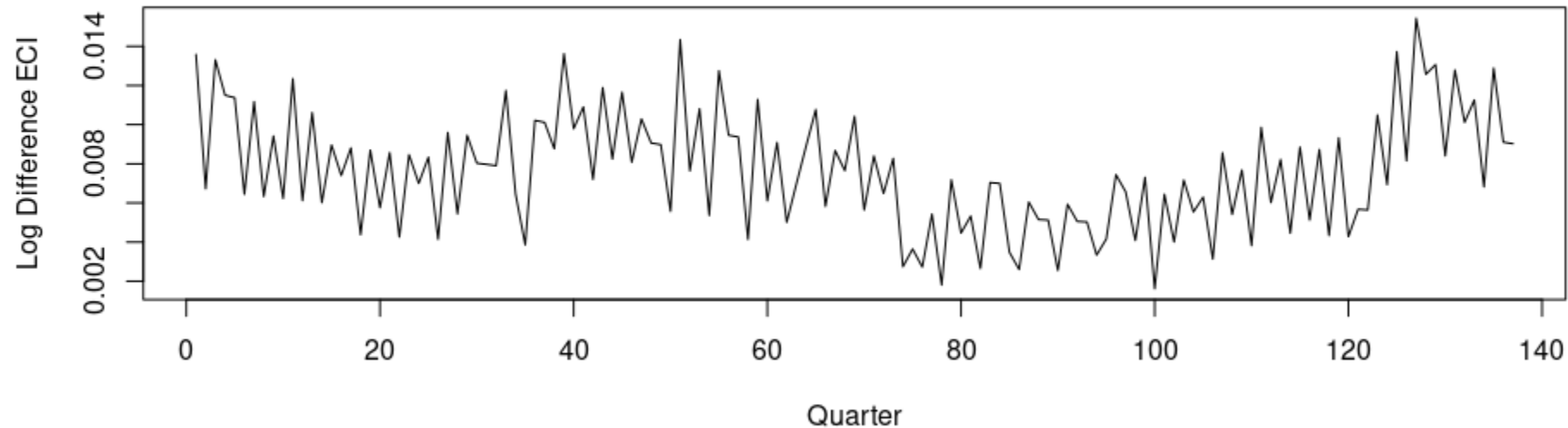


WAGE INFLATION MODEL SELECTION

3. Simplifying the Model



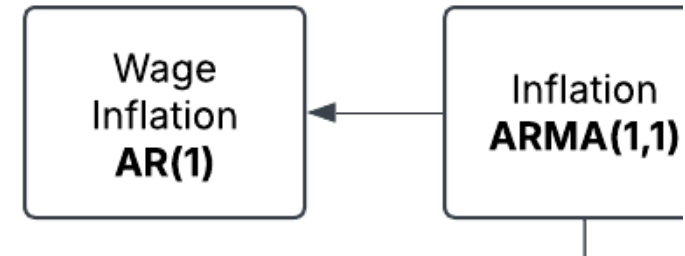
ECI Time-Series



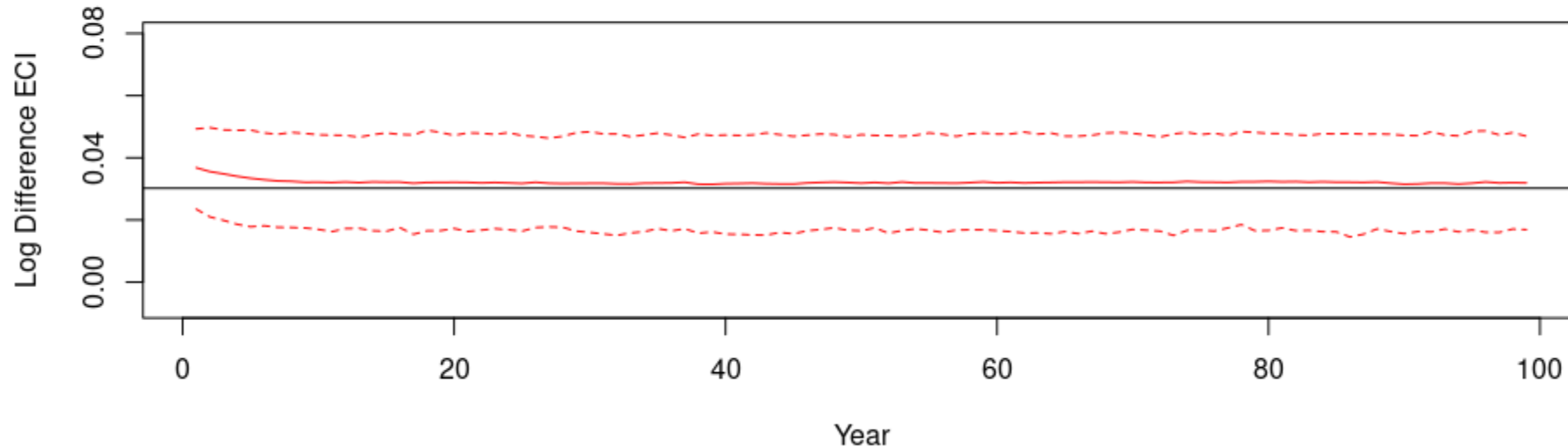
WAGE INFLATION MODEL SELECTION

3. Simplifying the Model

- Yearly instead of Quarterly
- CPI instead of lagged CPI as predictor
- AR(1)



Yearly ECI Draws AR(1) w/ CPI as Predictor



INFLATION MODELS

1. General Inflation: ARMA(1,1)

$$\log \left(\frac{\text{CPI}_t}{\text{CPI}_{t-1}} \right) = q_t = \mu + \phi q_{t-1} + \theta \epsilon_{t-1} + \epsilon_t, \quad \text{where } \epsilon_t \sim N(0, \sigma^2).$$

2. Medical Inflation: ARMA(1,1)

$$\log \left(\frac{\text{MedCPI}_t}{\text{MedCPI}_{t-1}} \right) = m_t = \mu + \phi m_{t-1} + \theta \epsilon_{t-1} + \epsilon_t, \quad \text{where } \epsilon_t \sim N(0, \sigma^2).$$

3. Wage Inflation: AR(1) w/ CPI as predictor

$$\log \left(\frac{\text{ECI}_t}{\text{ECI}_{t-1}} \right) = w_t = \mu + q_t + \phi w_{t-1} + \epsilon_t, \quad \text{where } \epsilon_t \sim N(0, \sigma^2).$$

INTEREST RATES

1. Interest Rate Transformation:

For $i = \{3/12, 1, 2, 3, 5, 7, 10, 20, 30\}$:

$$\tilde{r}_{i,t} = \begin{cases} r_{i,t} & \text{if } r_{i,t} > \bar{r} \\ \bar{r} - \bar{r} \log(\bar{r}) + \bar{r} \log(r_{i,t}) & \text{if } r_{i,t} \leq \bar{r} \end{cases},$$

where $\bar{r} = .005$ (Bégin 2021)

2. Short Term Rates: ARMA(1,1) & GARCH(1,1)

$$\tilde{r}_{3/12,t} = \mu + q_t + \phi \tilde{r}_{3/12,t-1} + \theta \epsilon_{t-1} + \epsilon_t, \quad \text{where } \epsilon_t \sim \mathcal{N}(0, \sigma_t^2),$$

and

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2,$$

YIELD CURVES

1. Functional Form

Level: $\tilde{r}_{3/12,t}$

Slope: $\tilde{r}_{30,t} - \tilde{r}_{3/12,t}$,

Curvature: $\tilde{r}_{3/12,t} + \tilde{r}_{30,t} - 2\tilde{r}_{10,t}$.

2. Modeling Slope and Curvature: VAR(1)

$$\mathbf{F}_t = \boldsymbol{\mu} + \mathbf{A} \mathbf{F}_{t-1} + \mathbf{b} \tilde{r}_{3/12,t} + \boldsymbol{\epsilon}_t, \quad \text{where } \boldsymbol{\epsilon}_t \sim N_2(\mathbf{0}, \boldsymbol{\Sigma}),$$

3. Yield Curve Construction: Factor Loadings

$$\tilde{r}_i = \mathbf{x}' \boldsymbol{\beta}_i + \epsilon_i, \quad i = \{1, 2, 3, 5, 7, 20\}, \quad \text{where } \epsilon_i \sim N(0, \sigma_i^2).$$

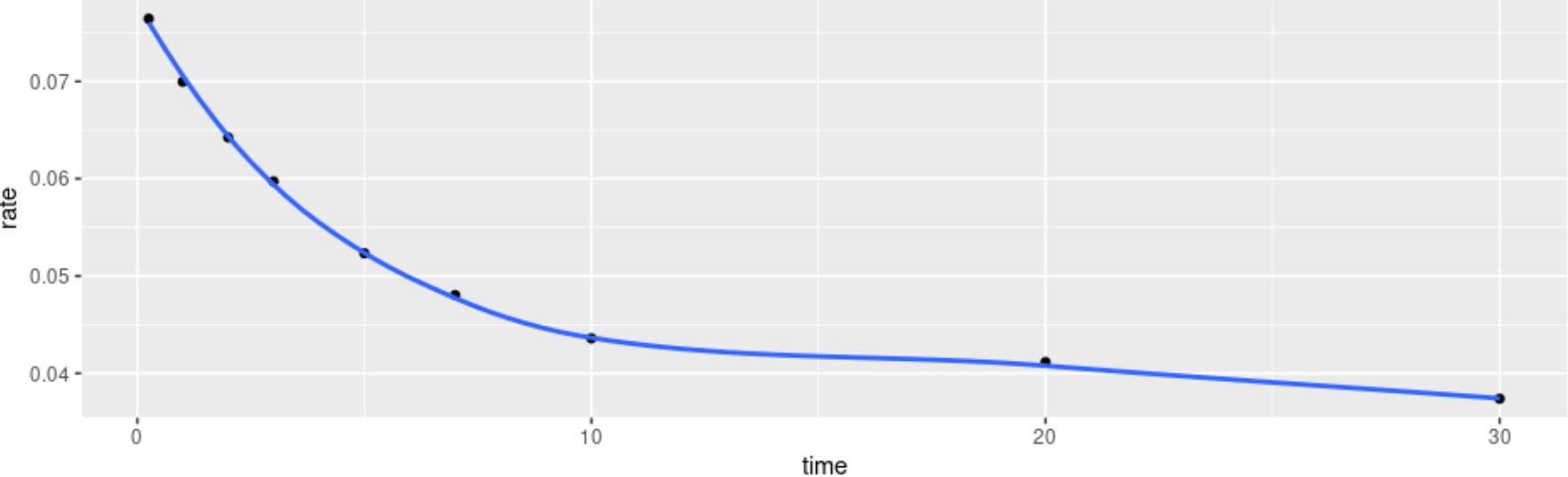
$$\mathbf{x}' = [\text{level}, \text{slope}, \text{curvature}]$$

YIELD CURVES



Sample Yield Curve

Slope: -0.0390571054597267. Curve: 0.0266429314273791



EQUITY RETURNS

- Broad Based Index Fund: S&P 500
- Regime Switching GARCH(1,1):

$$y_t = \mu + q_t \gamma_1 + r_{3/12,t} \gamma_2 + \epsilon_t, \quad \text{where } \epsilon_t \sim \mathcal{N}(0, \sigma_t^2),$$

and

$$\sigma_t^2 = \omega_{R_t} + \alpha_{R_t} \epsilon_{t-1}^2 + \beta_{R_t} \sigma_{t-1}^2$$

Transition Matrix: $\Theta = \begin{pmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{pmatrix}$

SIMULATION STEPS:

1. Generate n simulations from the ESG x years into the future
2. Draw n sample death ages from mortality table
3. Apply simulations to a financial plan
4. Calculate success criteria

CASE STUDY

Primary Question: For a given amount of spending, how much should be annuitized?

Key Parameters:

1. Withdrawal Rate: annual spending as a percentage of savings at retirement
2. Annuity Proportion: percentage of savings at retirement to annuitize

Key Result: Probability of out-living assets

WITHDRAWAL RATES AND ANNUITY PRICING

1. Withdrawal Rate: r

$$r = \frac{\text{Yearly Expenses}}{\text{Retirement Savings}}$$

$$\text{Monthly Expenses} = c = \text{Retirement Savings} \times \frac{r}{12}.$$

2. Annuity Payout: b

$$p_x = (1 + l) \sum_{t=1}^T (1 + r_{t/12})^{-t/12} K_{x+\frac{t}{12}},$$

$$\text{Monthly Annuity Payout} = b = \frac{\alpha \times \text{Retirement Savings}}{p_x}$$

RETIREMENT PLAN SET-UP

- Stock Market withdrawal amount:

$$\text{CPI}_t c - b$$

- Portfolio Value at time t :

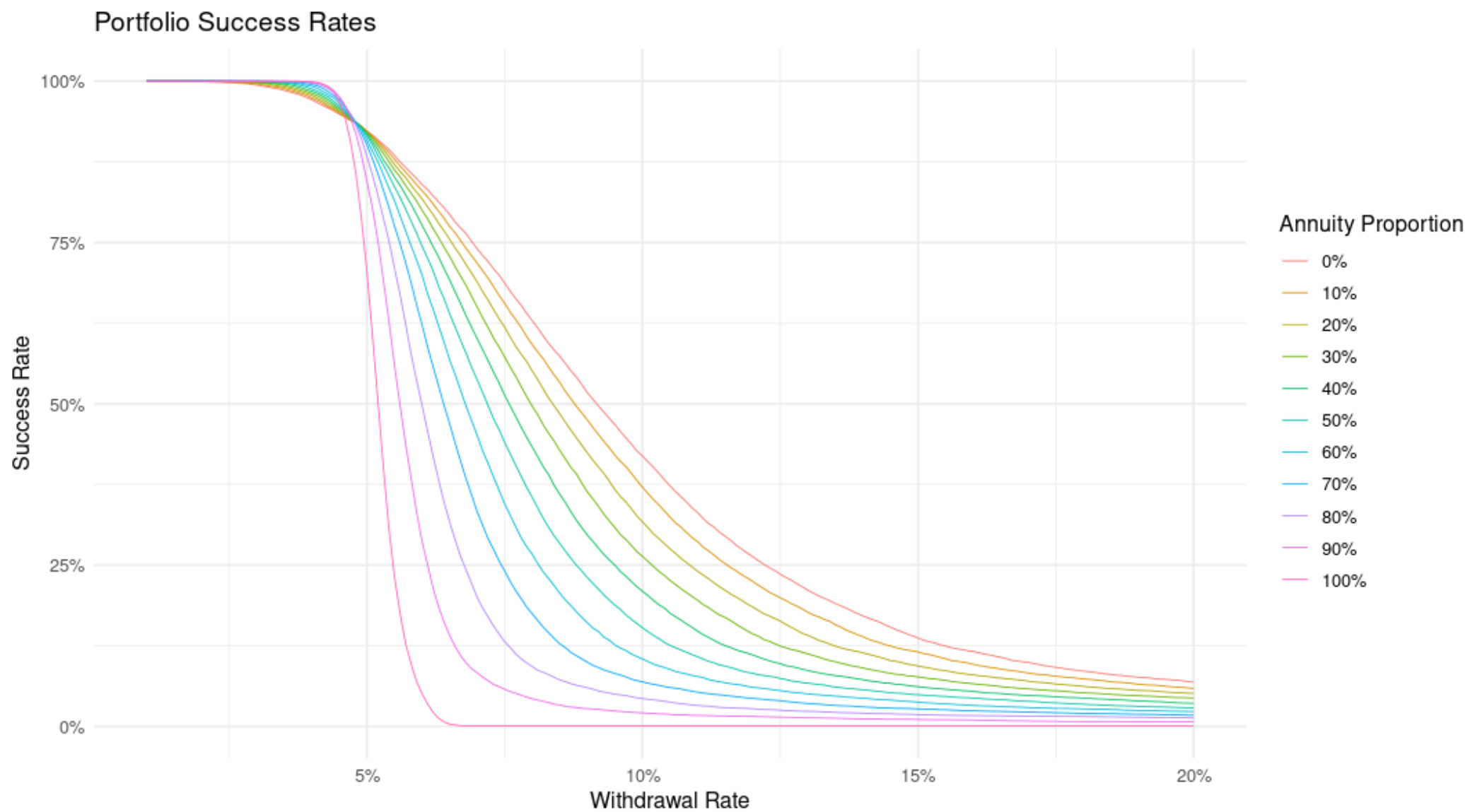
$$v_t = (v_{t-1} - (\text{CPI}_t c - b)) y_t$$

- Final Set-up:

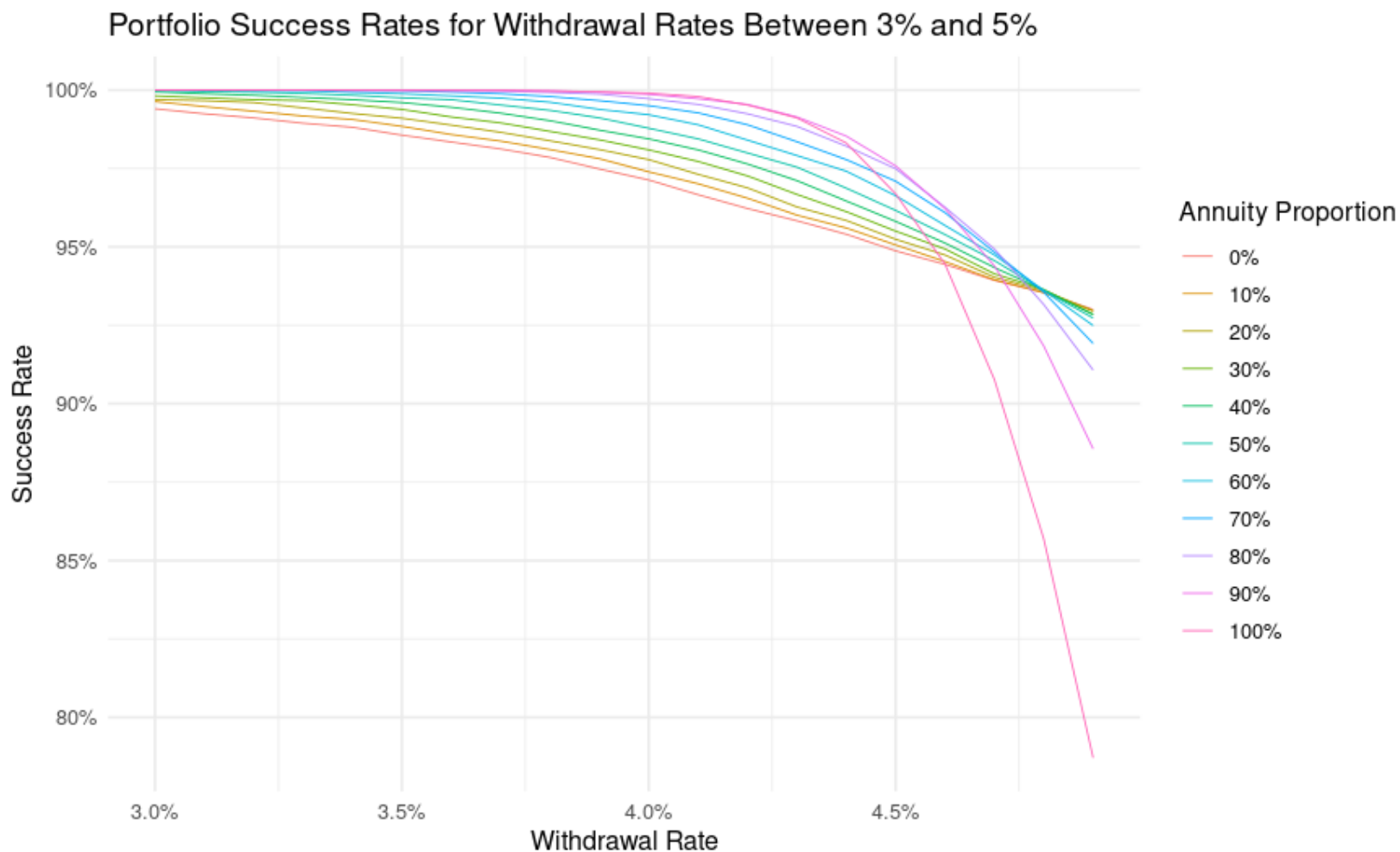
$$v_t = \left(v_{t-1} - \left(\underbrace{(0.8 \text{CPI}_t + 0.2 \text{MedCPI}_t) c}_{\text{weighted spending}} - \underbrace{b}_{\text{annuity payout}} \right) \right) \underbrace{y_t}_{\text{equity returns}}$$

- Success Criteria: $v_t > 0$ at time of death

RESULTS (10,000 SIMULATIONS)



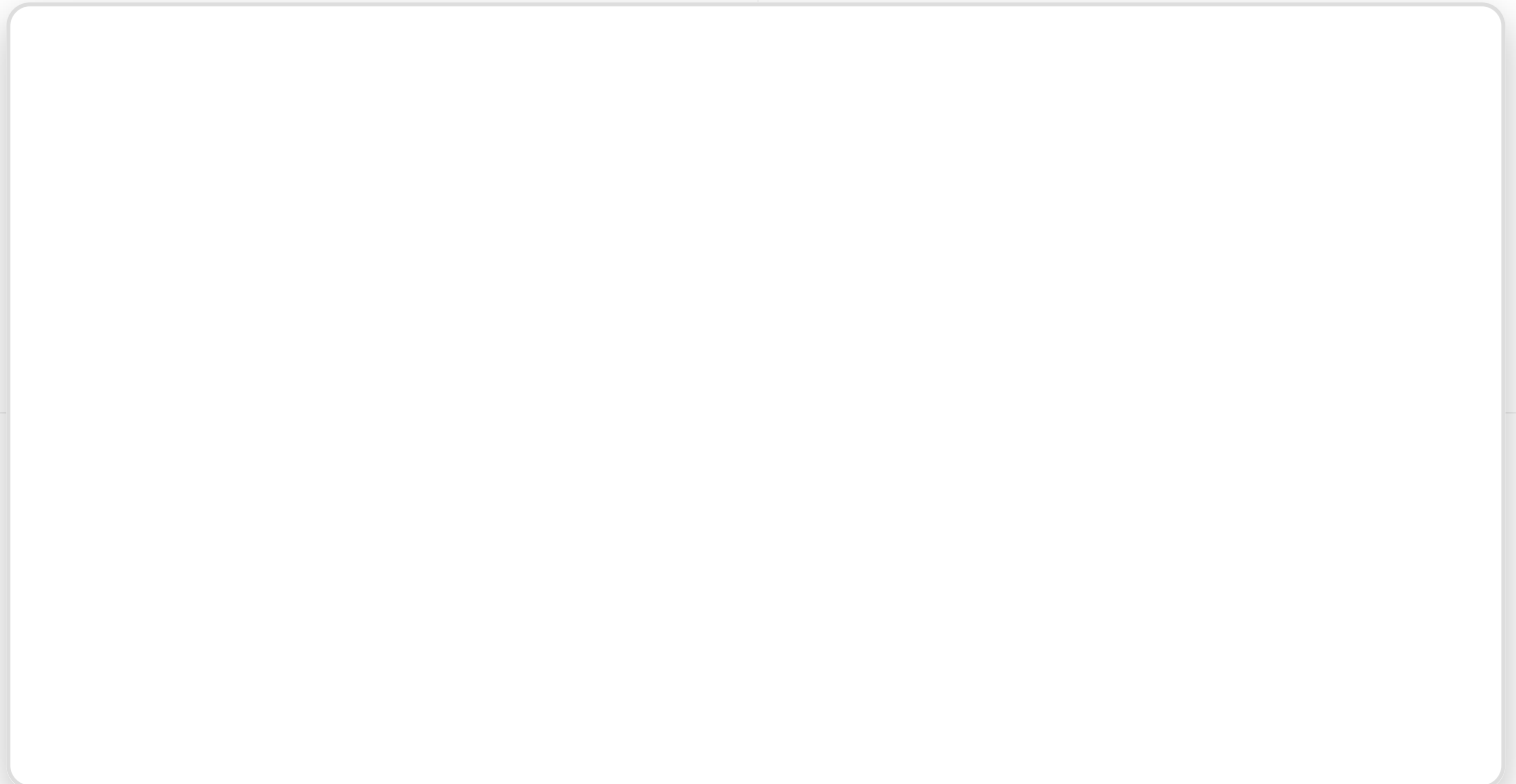
RESULTS (10,000 SIMULATIONS)



- 4% Rule?

LOOKING AHEAD

- Implement more realistic retirement plans
- Pre-retirement scenarios (Wage-inflation)
- Shiny App



THANK YOU